2f. Content of Discrete Mathematics (Optional paper Y544)

Introduction to Discrete Mathematics.

Discrete Mathematics is the part of mathematics dedicated to the study of discrete objects. Learners will study both pure mathematical structures and techniques and their application to solving real-world problems of existence, construction, enumeration and optimisation. Areas studied include counting, graphs and networks, algorithms, critical path analysis, linear programming and game theory.

7.01 Mathematical Preliminaries
Learners are introduced to the fundamental categorisation of problems as existence, construction, enumeration and optimisation. They are also introduced to counting techniques that have a wide application across Discrete Mathematics.

7.02 Graphs and Networks
Graphs and networks are introduced as mathematical objects that can be used to model real world systems involving connections and relationships. The pure mathematics of graph theory is studied including isomorphism, and Eulerian, Hamiltonian and planar graphs.

7.03 Algorithms
The algorithmic approach to problem solving in introduced via sorting and packing problems. The run-time and order of an algorithm are studied, including the hierarchy of orders.

7.04 Network Algorithms
Problems involving networks are introduced: shortest path; minimum connector; travelling salesperson and route inspection.
Standard network algorithms are studied and used to solve these problems.

7.05 Decision Making in Project Management
Networks are applied to decision making, in particular to activity networks and critical path analysis, including scheduling.

7.06 Graphical Linear Programming
The concept of linear programming is explored as a tool for optimisation, including integer programming. Linear programmes are solved graphically.

7.07 The Simplex Algorithm
The simplex algorithm is used to solve optimisation problems.

7.08 Game Theory
Problems of conflict and cooperation are explored using game theory, including both pure and mixed strategies.

Assumed knowledge
Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. They are also assumed to know the content of Pure Core (Y540 and Y541). All of this content is assumed, but will only be explicitly assessed where it appears in this section.

The use of algorithms

Learners will only be expected to use specific algorithms if instructed to do so in the question. For example a list may be sorted by inspection unless a question specifically asks for the use of a sorting algorithm, and lengths of shortest paths may be found by inspection unless a question specifically asks for the use of Dijkstra's algorithm.

The Formulae Booklet contains sketches of some of the algorithms found in this area. The focus of the study of algorithms in this area should be on understanding the theory and processes, not on memorisation or practising the rote application of algorithms by hand. In the classroom, technology should be used to demonstrate the power of these algorithms in large scale problems; in the assessment, learners will be asked to demonstrate their understanding of the application of algorithms to small scale problems only.

Use of technology

To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. Graphing tools: Learners could use graphing software to perform graphical linear programming, and to investigate the effects on the solution of changing coefficients and parameters.

2. Networks and network algorithms: Learners could use software to investigate networks and to implement network algorithms, in particular for networks which are too large to work with by hand.

3. Algorithms: Learners could use spreadsheets or a suitable programming language to implement simple algorithms and to understand how to create algorithms to perform simple tasks.

4. Computer Algebra System (CAS): Learners could use CAS software to draw and manipulate graphs, to explore algebraic relationships. This is best done in conjunction with other software such as graphing tools and spreadsheets.

5. Simulation: Learners could use spreadsheets to simulate contexts in game theory, including investigating the long term effects of particular strategies.

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When this course is being co-taught with AS Level Further Mathematics A (H235) the ‘Stage 1’ column indicates the common content between the two specifications and the ‘Stage 2’ column indicates content which is particular to this specification. In each section the OCR reference lists ‘Stage 1’ statements before ‘Stage 2‘ statements.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01 Mathematical Preliminaries			
7.01a	Types of problem	a) Understand and be able to use the terms “existence’” “construction”, “enumeration” and “optimisation” in the context of problem solving. *Includes classifying a given problem into one or more of these categories.*	
7.01b	Set notation	b) Understand and be able to use the basic language and notation of sets. *Includes the term “partition” and counting the number of partitions of a set including with constraints.*	
7.01c	The pigeonhole principle	c) Be able to use the pigeonhole principle in solving problems.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01d	Arrangement and selection problems	d) Understand and use the multiplicative principle. Includes knowing that the number of arrangements of n distinct objects is $\prod_{r=1}^{n} r = n!$.	
7.01e		e) Be able to enumerate the number of ways of obtaining an ordered subset (permutation) of r elements from a set of n distinct elements. *Includes using the notation* ${}_n\mathrm{P}_r = {}^n\mathrm{P}_r$.	
7.01f		f) Be able to enumerate the number of ways of obtaining an unordered subset (combination) of r elements from a set of n distinct elements. *Includes using the notation* ${}_n\mathrm{C}_r = {}^n\mathrm{C}_r$.	
7.01g 7.01h		g) Be able to solve problems about enumerating the number of arrangements of objects in a line, including those involving: 1. repetition, *e.g. how many different eight digit numbers can be made from the digits of* 12333210*?* 2. restriction, *e.g. how many different eight digit numbers can be made from the digits of* 12333210 *if the two* 2*s cannot be next to each other?*	h) Be able to solve problems about enumerating the number of arrangements of only some of a group of objects. *e.g. how many different four digit numbers can be made from the digits of* 12333210*?* *e.g. how many different numbers greater than* $20\,000$ *can be made from the digits of* 12333210*?*
7.01i 7.01j		i) Be able to solve problems about selections, including with constraints. *e.g. Find the number of ways in which a team of* 3 *men and* 2 *women can be selected from a group of* 6 *men and* 5 *women.*	j) Be able to solve problems with several constraints. *e.g. Given a graph showing who dislikes who, find the number of ways of choosing* 3 *men and* 2 *women from a group of* 4 *men and* 4 *women so that no two people chosen dislike each other.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01k 7.01l	The inclusion-exclusion principle	k) Be able to use the inclusion-exclusion principle for two sets in solving problems. *e.g.* $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. *Venn diagrams may be used.* *e.g. How many integers in* $\{1, 2, ..., 100\}$ *are not divisible by* 2 *or* 3*?*	l) Be able to extend the inclusion-exclusion principle to more than two sets. *e.g.* $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B)$ $- n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$. *Venn diagrams may be used.* *e.g. How many integers in* $\{1, 2, ..., 100\}$ *are not divisible by* 2, 3 *or* 5*?*
7.01m			m) Be able to find derangements. *Includes enumeration of the number of derangements of* n *objects,* D_n*, by ad hoc methods only.*